SPEC IS COVERED BY THE FIBRE OVER V(F) AND THE STANDARD OPEN D(F) FOR A MAP IDENTIFYING R/FR

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Allowed standard prerequisites (editorial).

- Spec(R) decomposes as the disjoint union of the closed subset V(f) = Spec(R/fR) and the standard open D(f) = Spec(R_f). invoked at proof lines 25179-25184. Splits the target into the two pieces to be covered.

Source reference. Original section 00DY: sources/stacks-chow/tags.txt, source lines 515–515. Cited at proof lines 25180-25181 for the background on Spec; the section is used inside the counted span.

Source reference. Original lemma 00E5: sources/stacks-chow/tags.txt, source lines 522–522. Cited at proof lines 25181-25182 for V(f) = Spec(R/fR); the conclusion is used inside the counted span.

Source reference. Original lemma 00E4: sources/stacks-chow/tags.txt, source lines 521–521. Cited at proof lines 25181-25182 for D(f) = Spec(R_f); the conclusion is used inside the counted span.

Source reference. Original lemma 0BNJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25129–25129. Cited at proof lines 25185-25186 to identify the completion case as a special case; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring, let $f \in R$, and let $R \to R'$ be a ring map which induces an isomorphism $R/fR \to R'/fR'$. The map $\mathrm{Spec}(R') \amalg \mathrm{Spec}(R_f) \to \mathrm{Spec}(R)$ is surjective. For example, the map $\mathrm{Spec}(R^\wedge) \amalg \mathrm{Spec}(R_f) \to \mathrm{Spec}(R)$ is surjective.*

Proof. Recall that $\mathrm{Spec}(R) = V(f) \amalg D(f)$ where $V(f) = \mathrm{Spec}(R/fR)$ and $D(f) = \mathrm{Spec}(R_f)$, see Algebra, Section 00DY and especially Lemmas 00E5 and 00E4. Thus the lemma follows as the map $R \to R/fR$ factors through R'. The last statement is a special case of the first statement by Lemma 0BNJ. □

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